# Palaeomagnetism of Late Precambrian or Cambrian diorites from Leicestershire, U.K.

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*Summary.* Af demagnetization of the Late Precambrian or Cambrian diorites of Leicestershire reveals at least two components of magnetization in sites with stable, coherent NRM. A high remanent-coercivity component yields a palaeomagnetic pole at 141° E 63° N ($A_{95}$ = 9°) in agreement with results from other Late Precambrian or Cambrian rocks in North Wales and Norway. A softer component may be of recent viscous origin, or have been acquired during Cambrian remagnetization.

## 1. Introduction

Dioritic and tonalitic rocks are exposed below Keuper Marl Triassic cover in several Leicestershire quarries located W of Leicester, near the M1 Motorway (Fig. 1). At Enderby the diorites intrude slates probably of Precambrian to Cambrian age (Hill & Bonney, 1878; Cribb, 1975); elsewhere the intrusive relationships of the diorities below the Triassic cover are not apparent. Rb–Sr whole rock ages of 546 ± 22 Ma and 552 ± 58 Ma ($2\sigma$) from the diorites of south Leicestershire (Enderby–Croft region, Fig. 1), and from S Charnwood (Groby–Markfield region) respectively indicate a Late Precambrian to Cam-

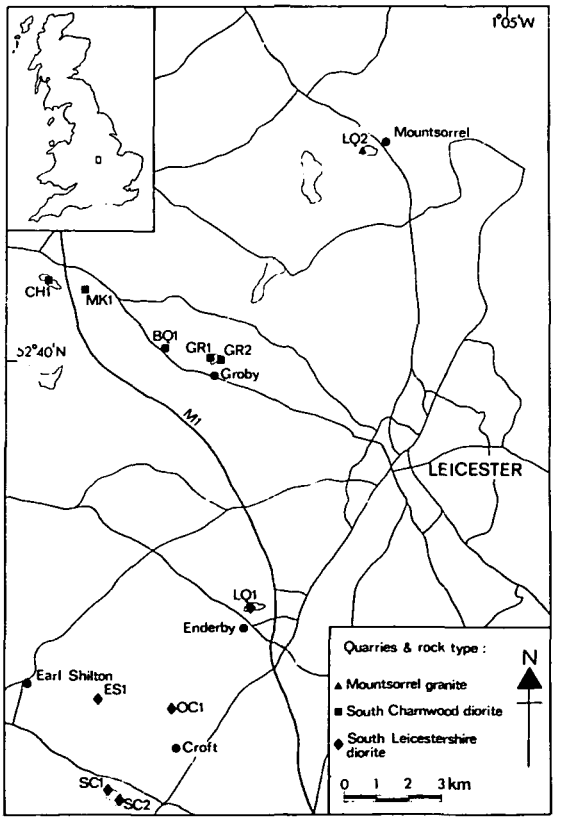



Figure 1. Location of palaeomagnetic sites in quarries of the Leicestershire diorites.

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Table 1. Palaeomagnetic results for the Leicestershire diorites

| Site | Region and rock type | Locality | D | I | N | R | K | $a_{95}$ |
|---|---|---|---|---|---|---|---|---|
| | | Total NRM | | | | | | |
| LQ2* | Mountsorrel granite | Quarry at Buddon Hill | 64 | 38 | 4 | 3.0 | 3 | 66 |
| BQ | South Charnwood diorites | Bradgate quarry | 12 | 57 | 10 | 9.5 | 18 | 12 |
| GR1† | South Charnwood diorites | Sheethedges quarry, Groby | 58 | 73 | 7 | 6.9 | 55 | 8 |
| GR2 | South Charnwood diorities | Quarry, Groby | 34 | 79 | 6 | 5.8 | 26 | 13 |
| CH* | South Charnwood diorities | Cliffe Hill | 356 | 64 | 6 | 5.8 | 26 | 13 |
| MK | South Charnwood diorities | Markfield quarry | 358 | 56 | 5 | 4.9 | 38 | 13 |
| LQ1 | South Leicestershire diorities | Enderby quarry | 8 | 48 | 9 | 7.3 | 5 | 27 |
| OC | South Leicestershire diorities | Old Croft quarry | 351 | 64 | 8 | 4.3 | 2 | 57 |
| ES* | South Leicestershire diorities | Quarry SE of Earl Shilton | 13 | 79 | 6 | 5.8 | 31 | 12 |
| SC1† | South Leicestershire diorities | Stoney Cove | 32 | −41 | 6 | 1.5 | 1 | 180 |
| SC2† | South Leicestershire diorities | Stoney Cove | 158 | 73 | 4 | 3.3 | 5 | 49 |
| | | Af cleaned NRM | | | | | | |
| BQ | | | 21 | 41 | 8 | 7.7 | 51 | 11 |
| MK | | | 15 | 53 | 5 | 4.9 | 48 | 11 |
| LQ1 | | | 18 | 51 | 8 | 8.0 | 272 | 3 |
| GR2 | | | 33 | 78 | 6 | 5.9 | 51 | 9 |
| OC | | | 292 | 72 | 7 | 6.9 | 49 | 9 |
| | | Mean of BQ, MK and LQ1 | | | | | | |
| 3 sites | | | 18 | 48 | 3 | 3.0 | 136 | 11 |
| | | Palaeomagnetic pole 141° E 63° N ($A_{95}$ = 9°) | | | | | | |
| 21 specimens | | | 19 | 48 | 21 | 20.5 | 43 | 5 |
| | | Palaeomagnetic pole 141° E 63° N ($d\psi$ = 4, $d\chi$ = 6) | | | | | | |

Notes: * Sites with unstable NRM, omitted from overall analysis; † sites with stable NRM but scattered directions, omitted from overall analysis.

D and I are the mean declination and inclination of the mean remanence vector in N sites or specimens; R is the resultant of N unit vectors; K is Fisher's (1953) precision parameter; $a_{95}$ is the semi-angle of the cone of 95% confidence about the mean direction; $A_{95}$ is the semi-angle of the cone of 95% confidence about the mean of virtual geomagnatic poles; $d\psi$ and $d\chi$ are the semi-axes of the 95% confidence oval about the mean pole.

brian age of emplacement (Cribb, 1975). The ages of the diorites in these two regions are not significantly different and suggest that the quarries expose different parts of a single large dioritic complex to the W of Leicester (Cribb, 1975). An Rb–Sr whole rock age of $433 \pm 17$ Ma for the Mountsorrel granite (Cribb, 1975) implies that it is distinct from the diorites.

Palaeomagnetic collections were made at 10 sites where the Charnwood S and S Leicestershire diorites have been exposed by quarrying (Fig. 1; Table 1). Quarries near Croft and Groby were sufficiently large to allow sampling at two sites. One site was sampled in the large quarry in Mountsorrel granite N of Leicester.

## 2. NRM and af demagnetization

Intensities of natural remanent magnetization (NRM) of the diorites range from about $10^{-3}$ $Am^{-1}$ to about 1 $Am^{-1}$ and are very variable within and between sites. The 11 site mean directions of total NRM (Table 1) are grouped about an overall mean direction, D = 24°, I = 62° ($a_{95}$ = 22°), which is not significantly different at the 95% probability level from the present field direction for Leicestershire (D = 352°, I = 68°).

Of the 11 sites only eight diorite sites are stable against alternating field (af) demagne-

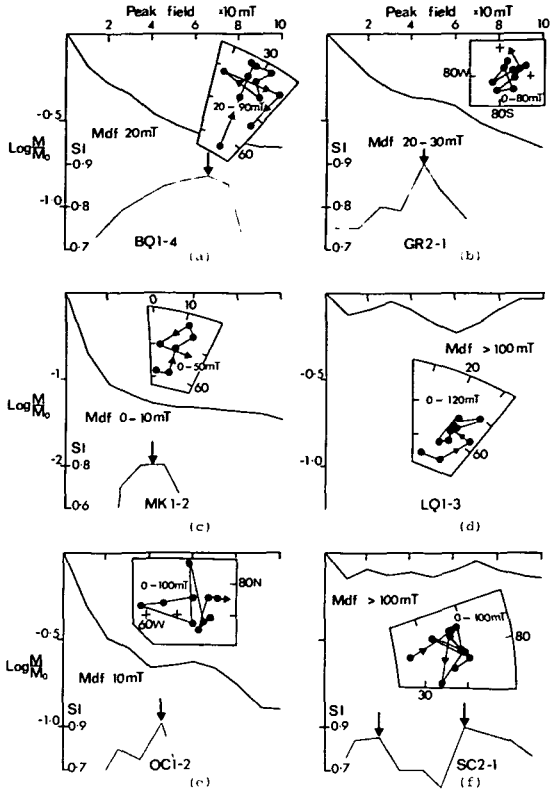



Figure 2. Progressive af demagnetization of NRM in the Leicestershire diorites, showing corresponding directional changes (all directions downward) and variation of stability indices (Briden, 1972) with peak field. Median destructive fields (Mdf) are also given.

tization and of these three (GR1, SC1 and SC2) have large within-site scatters of NRM directions after optimum field demagnetization. The NRM of the Mountsorrel granite and of the diorite from quarries at Cliffe Hill and near Earl Shilton was almost entirely demagnetized in peak fields less than about 20 mT, and showed large erratic directional changes.

Three of the five stable and well-grouped sites gave similar endpoints with intermediate inclinations, oblique to the direction of the present field and total NRM (Fig. 2a, c and d). Planar rotation of the resultant NRM from near the present field direction toward the stable endpoint during af demagnetization at low peak fields suggests the presence of a two-component NRM. Variability of the remanent-coercivity spectra of the soft and hard components of NRM between sites is reflected in the different NRM decay curves, and variable single-peaked distributions of Briden's (1972) stability index (Fig. 2a, c and d). Median destructive fields range from about 10 mT to greater than 100 mT (for site LQ).

The sites GR2 and OC have very steep endpoints after demagnetization in peak fields up to 100 mT (Fig. 2b and e). The variations of stability indices of these sites are similar, with subsidiary and major peaks at fields of about 25 mT and 45 mT respectively, which probably reflect both relatively soft and hard NRM components with similar directions.

The demagnetization behaviour of the NRM for site SC2 (Fig. 2f) is typical of that for

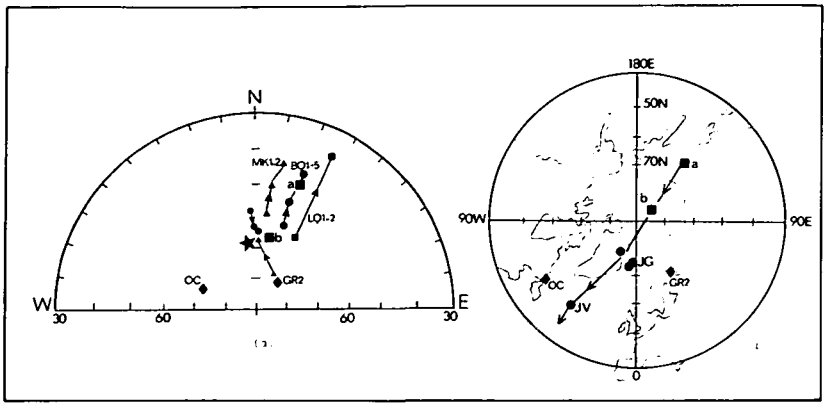



Figure 3. (a) Directional changes of NRM in sites MK, LQ1 and BQ on af demagnetization (large dots and triangles, small squares) and the corresponding removed vector directions (smaller dots and triangles). The latter converge toward the present field direction (star) and the mean total NRM direction (B) for the eight stable diorite sites. The mean direction of cleaned NRM for the three stable sites is 'A' (all directions downward, stereographic projection). (b) Position of the VGP corresponding to the total NRM or soft component of NRM in the diorites (B) on the polar path joining the early Cambrian pole A with the poles for the Jersey granites (JG) and volcanics (JV). Polar stereographic projection.

three sites which, although stable against af demagnetization, have scattered directions of NRM after optimum field cleaning. The reason for the scattered endpoints within sites is not clear, and these sites are excluded from further analysis.

## 3. Discussion

After optimum field demagnetization, the three stable and consistently orientated sites gave an overall mean direction of D = 18°, I = 48° ($a_{95}$ = 11°, Fig. 3) with a palaeomagnetic pole position at 141° E 63° N ($A_{95}$ = 9°), in close agreement with results from the early Cambrian Fen Complex of Norway (Poorter, 1972) and the Bangor red beds of North Wales (Duff, 1978, unpubl. thesis, Leeds Univ.). The NRM probably dates from Late Precambrian to early Cambrian intrusion of the Leicestershire diorites. A two-tier analysis (Watson & Irving, 1957) showed that the between-site scatter is not significantly different from the within-site scatter at the 95% probability level ($F_{4,36}$ = 2.4) for the three sites. This supports the interpretation from the Rb–Sr results that the diorites NW and SW of Leicester exposed by quarrying are all part of the same dioritic complex.

Convergence of the directions of removed vectors toward the present field direction during af demagnetization of the softer component in the three stable sites BQ, LQ1 and MK (Fig. 3a) implies that this component may be of recent viscous origin. However, it is also possible that the softer NRM component was acquired after early Cambrian intrusion but before middle Cambrian acquisition of NRM in Jersey (Channel Island) granites and volcanics, because the virtual pole which corresponds to the soft component in the stable diorite sites has an intermediate position on Duff's (1978, unpubl. thesis) apparent polar path for Europe between the pole (A), which corresponds to the early Cambrian component of the three stable sites, and the early to middle Cambrian poles for Jersey (Fig. 3b). Possible Cambrian remagnetization of the lower part of the remanent-coercivity spectra is supported by

evidence of Cambrian remagnetization (Lomax, 1975, unpubl. thesis, Leeds Univ.) and isotopic rehomogenization (e.g. Lambert & Rex, 1966) of Precambrian rock units in other regions of the Midlands and Welsh Borderlands.

*Acknowledgements.* This work was undertaken during tenure of a University of Leeds Studentship, with the encouragement of Professor J. C. Briden. Mr W. Iles assisted during field sampling.

## References

Briden, J. C. 1972. A stability index of remanent magnetization. *J. geophys. Res.* **77**, 1401–5.

Cribb, S. J. 1975. Rubidium–strontium ages and strontium isotope ratios from the igneous rocks of Leicestershire. *Jl geol. Soc. Lond.* **131**, 203–12.

Fisher, R. A. 1953. Dispersion on a sphere. *Proc. R. Soc. Lond.* A **217**, 295–305.

Hill, E. & Bonney,T. G. 1878. On the pre-Carboniferous rocks of Charnwood Forest. *Q. Jl geol. Soc. Lond.* **34**, 199–239.

Lambert, R. St J. & Rex, D. C. 1966. Isotopic ages of minerals from the Precambrian complex of the Malverns. *Nature, Lond.* **209**, 605–6.

Poorter, R. P. E. 1972. Preliminary palaeomagnetic results from the Fen Carbonatite complex, South Norway. *Earth Planet. Sci. Lett.* **17**, 194–8.

Watson, G. S. & Irving, E. 1957. Statistical methods in rock magnetism. *Mon. Not. Roy. Astron. Soc. Geophys. Supp.* **7**, 289–300.

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